

Lab 9

Programing with Python Lab

BCACC393

Dataset: UCI Machine Learning Repository — **Automobile Dataset** (car pricing dataset). The exercises below focus on **fundamental data analytics, data cleaning, descriptive statistics, grouping, correlation, and basic plots** using Python, Pandas, NumPy, and Matplotlib.

Dataset Link: <https://archive.ics.uci.edu/dataset/10/automobile>

Common dataset loading: The UCI Automobile dataset is commonly distributed as a CSV/text file with missing values represented by ?. In the solutions below, the file is assumed to be named `imports-85.data`, with the standard UCI column names.

1. Loading and Exploring the Dataset

Load the UCI Automobile dataset into a Pandas DataFrame. Display:

- First five records
- Last five records
- Number of rows and columns
- Column names
- Data types of all columns

2. Dataset Information and Summary Statistics

Use Pandas to examine the structure of the Automobile dataset. Display the dataset information and generate descriptive statistics for all numerical columns.

3. Handling Missing Values

The Automobile dataset contains missing values represented by ?. Replace these values with NaN and:

- Count missing values in each column.
- Display columns containing missing values.
- Display the total number of missing values in the dataset.

4. Cleaning the Price Column

Convert the price column into a numeric data type. Handle missing values appropriately and calculate:

- Minimum car price
- Maximum car price
- Average car price

- Median car price

5. Frequency of Car Makes

Find the number of cars belonging to each make. Display the **top 10 car manufacturers** based on the number of cars in the dataset.

6. Body Style Analysis

Find the frequency of different body-style categories such as sedan, hatchback, wagon, convertible, and hardtop. Display the results in descending order.

7. Fuel Type Analysis

Analyze the fuel-type column and determine:

- Number of cars using each fuel type.
- Percentage of cars using each fuel type.

Represent the result using a suitable **bar chart**.

8. Car Price Distribution

Calculate the basic statistics of car prices and create a **histogram** showing the distribution of car prices.

Add a suitable title, X-axis label, Y-axis label, and grid.

9. Average Price by Car Make

Calculate the **average car price for each manufacturer**. Display the results in descending order and identify the five most expensive manufacturers based on average price.

10. Average Price by Body Style

Calculate and compare the average price of cars for each body-style. Represent the results using a **bar chart**.

11. Horsepower Analysis

Analyze the horsepower column and calculate:

- Minimum horsepower
- Maximum horsepower
- Mean horsepower
- Median horsepower
- Standard deviation

Also create a histogram of horsepower.

12. Engine Size Analysis

Analyze the engine-size column and determine:

- Minimum engine size
- Maximum engine size
- Average engine size
- Median engine size

Create a histogram to visualize the distribution of engine sizes.

13. Horsepower vs Price

Investigate the relationship between horsepower and price. Create a **scatter plot** with horsepower on the X-axis and price on the Y-axis.

Based on the plot, comment on whether there appears to be a relationship between horsepower and car price.

14. Engine Size vs Price

Create a scatter plot showing the relationship between engine-size and price. Calculate the correlation coefficient between the two variables.

15. Curb Weight vs Price

Analyze the relationship between curb-weight and price. Create a scatter plot and calculate the correlation coefficient.

16. Highway MPG Analysis

Analyze the highway-mpg column. Calculate its minimum, maximum, mean, and median values. Create a histogram showing the distribution of highway mileage.

17. City MPG vs Highway MPG

Create a scatter plot comparing city-mpg and highway-mpg. Calculate the correlation coefficient between the two variables.

Determine whether cars with higher city mileage generally also have higher highway mileage.

18. Number of Doors by Body Style

Analyze the relationship between num-of-doors and body-style. Create a suitable frequency table and visualize the result using a **bar chart**.

19. Fuel Type and Average Price

Calculate the average price of cars for each fuel-type. Compare the prices of gasoline and diesel cars using a bar chart.

20. Drive Wheel Analysis: Analyze the drive-wheels column and determine the number of cars having:

- Front-wheel drive (fwd)
- Rear-wheel drive (rwd)
- Four-wheel drive (4wd)

Create a bar chart to visualize the distribution.

21. Price Comparison by Drive Wheels

Calculate the average price of cars for each drive-wheels category. Display the result and create a bar chart comparing the average prices.

22. Correlation Analysis of Numerical Variables

Select important numerical variables such as:

- price
- engine-size
- horsepower
- curb-weight
- city-mpg
- highway-mpg

Calculate their correlation matrix and identify:

- The variable most strongly positively correlated with price.
- The variable most strongly negatively correlated with price.

23. Basic Car Price Analytics Report

Perform a complete basic analysis of the Automobile dataset. Your analysis should include:

1. Dataset dimensions and basic information.
2. Missing-value analysis.
3. Descriptive statistics of important numerical variables.
4. Top five car manufacturers by average price.
5. Most common body style.
6. Average price by fuel type.
7. Average price by drive wheels.
8. Histogram of car prices.
9. Scatter plot of horsepower vs price.
10. Scatter plot of engine size vs price.

Write a short conclusion summarizing the major observations from the analysis.